

SCSIT

IV Semester

Database

Exc. 6

CRUD operations

SQL create, insert, update, delete

CRUD operations

- ☐ Object creation
 - ☐ **CREATE**
 - ☐ Data retrieval
 - ☐ **SELECT**
 - ☐ Permanent data insertion
 - ☐ **INSERT INTO**
 - ☐ Permanent data change
 - ☐ **UPDATE**
 - ☐ Permanent data removal
 - ☐ **DELETE**
-

Table work_place

Id_place	Title	Salary	Sector	Phone	Vehicle	LAPTOP
1	Manager	45000	Marketing	1	1	1
2	DBA	27000	IT	0	0	0
3	Analyst	30000	IT	1	0	1
4	Programmer	25000	IT	0	0	0
5	Secretary	10000	Administration	0	0	0
6	Doorman	7000	Administration	0	0	0
7	Accountant	12000	Finance	1	0	0
8	Treasurer	13000	Finance	1	0	0
9	Driver	9000	Service	1	1	0
10	Event manager	23000	Marketing	1	1	1

CREATE operation 1/3

- ☐ Create new Database
ex. CREATE DATABASE UACS_test
- ☐ Create new data structure (**Table**, View, Function, Procedure, ...)

```
CREATE TABLE [TableName] (  
    [Col1] [Data type 1],  
    [Col2] [Data type 2],  
    . . .  
    [ColN] [Data type N])
```

CREATE operation 2/3

□ Example

```
CREATE TABLE [Customer](  
    [Id] [int] NOT NULL,  
    [Name] [nvarchar](100) NOT NULL,  
    [City] [nvarchar](100) NULL)
```

CREATE operation 2/3

□ Example

```
CREATE TABLE [Customer](  
    [Id] [int] NOT NULL,  
    [Name] [nvarchar](100) NOT NULL,  
    [City] [nvarchar](100) NULL)
```

CREATE operation 2/3

- ❑ Primary key concept
 - Short, Not changeable, Incremental, Unique
- ❑ Identity concept

```
CREATE TABLE [Customer1](  
  [Id] [int] IDENTITY(1,1) NOT NULL,  
  [Name] [nvarchar](100) NOT NULL,  
  [City] [nvarchar](100) NULL,  
  CONSTRAINT [PK_Customer] PRIMARY KEY CLUSTERED  
  (  
    [Id] ASC  
  )  
GO
```

INSERT

Inserts ONE row in a table:

INSERT INTO *table* (column1 [,column2..])

VALUES (*value1* [,*value2* ...]);

value $\mathbf{I}$ = value from the domain of the suitable attribute (column $\mathbf{I}$)

For the columns that don't have values in the specific INSERT query:

1. *Their names are written in the column part, but instead of some value, in the Value part, you assign **NULL** to them*
 2. *You don't write these columns in the INSERT statement at all.*
-

INSERT

Insert new data

```
INSERT INTO [TableName] ([Col1],[Col2],...)  
VALUES (Value1,Value2,...)
```

Example1:

```
INSERT INTO [Customer] ([Id], [Name], [City])  
VALUES (1, 'Zito Skopje', 'Skopje')
```

```
INSERT INTO [Customer] ([Id], [Name], [City])  
VALUES (2, 'Zito Strumica', 'Strumica')
```

Example2:

```
INSERT INTO [Customer] ([Id], [Name], [City])  
VALUES (3, 'Zito Veles', 'Veles'),  
       (4, 'Zito Bitola', 'Bitola')
```

Ass. 1.

□ In table

work_place(id_place,title,salary,sector,phone,vehicle,laptop)

we will insert new touples with these values:

1. 11, 'Security',16000,'Administration', 1,0,0 (Ass. Will do it)

Each student should insert

His/her IT 'Student', no data for salary,
'SCSIT', 1/0 i.e. weather he/she has phone or not 1/0 i.e.
weather he/she has vehicle or not,
1/0 i.e. weather he/she has laptop or not.

Solutions

1. INSERT INTO

work_place(id_place,title,salary,sector,phone,
vehicle,laptop)

VALUES

(11,'Security','Administration',16000, 1 ,0, 0)

2. ?

READ operation

- ▶ Read all data in the table

```
SELECT * FROM Customer
```

- ▶ Read only specific columns:

```
SELECT ID, Name, City  
FROM Customer
```

- ▶ Read specific columns and Rows:

```
SELECT ID, Name, City  
FROM Customer  
WHERE City = 'Skopje'
```

INSERT / SELECT

□ Example

INSERT INTO Customer1 (Name, City)

SELECT Name, City

FROM Customer

SELECT * FROM Customer1

SELECT * INTO Customer1_bck

FROM Customer1

UPDATE

Makes change of specific column field value. The change can be done on none, one or several rows at the same time.

UPDATE *table*

SET *column* = new_value1 [, *column2* = new_value2...]

[WHERE *condition*];

column_i is the name of the column whose field values) are to be changed
New_value_i is the new value that will be assigned to the column field(s)
after SQL query exec.

condition is statement that determines the table rows for update .

Ass. 2 Make updates

- 1. Give phones to all work places in IT sector.**
- 2. Work place Secretary should have salary of 12500 and a vehicle**
- 3. Reduce the salary for work places that have phone. Their salary should be 1500 less than it is now.**
- 4. To all work places with salary above 30000, the sector should be changed into VIP, but only if they have LAPTOP, phone and vehicle.**
- 5. Let all SCSIT students get LAPTOP and Vehicle 😊**

Answers

1. UPDATE WORK_PLACE
SET PHONE=1
WHERE SECTOR='IT'

CAN YOU TELL THE NUMBER OF TOUPLE THE CHANGE IS
MADE ON?

- 2. ?
- 3. ?
- 4. ?
- 5. ?

DELETE

Statements for whole DB Table row(s) permanent removal

DELETE FROM *table* [**WHERE** *condition*];

table is the table name

condition is statement that picks table rows that should be deleted.

These rows are PERMANENTLY deleted i.e. you can not restore their data back

With **delete from *table*, ALL** table rows will be deleted.

Ass. 3. Delete queries

1. Delete all data for work places in Administration
2. Delete data for work place with title "Outdoor consultant"
3. Delete work place with salary above 50000
4. Delete all data for SCSIT students
5. Delete all data from the table

Answers

1. DELETE FROM WORK_PLACE WHERE
SECTOR='Administration'

Can you tell me the number of deleted rows?

2. ?

3. ?

4. ?

5.?
